Complex numbers

1 A complex number

2 $0 + bi$ or bi

3

a $\sqrt{-1}$

b -1

4

a 0

b $-\sqrt{10}$

c Nonreal complex/pure imaginary

5

a $\sqrt{5}i$

b $\sqrt{14}$

Operations with complex numbers

6 $9 + 2i$

7 No, $-4i$

8 No, $\frac{6 \pm 36i}{2} = 3 \pm 18i$

9

a $-10i$

b $3i\sqrt{2}$

c $9 - 8i$

d $2 + 5i$

10 -8

11

a -19

b 5

c $\sqrt{5}$

d $\frac{1}{3}$

12

a $10 + 9i$ b $3 + 9i$ c $-3 + 5i$ d $6 - 11i$

e $-3.3 + 2.6i$ f $-11 + 16i$ g $-11 - 5i$ h $12 - 6i$

i $\frac{7}{6} - \frac{i}{20}$

13

a $-18 + 30i$ b $14 + 21i$ c $-10 + 8\sqrt{10}i$ d $96 - 16i$

e 65 f -29 g $-21 - 20i$ h $-\frac{1}{4} + \frac{3}{4}i$

i $16i + 12$ j $-14i - 48$

14

a

i -1 ii 1 iii -1 iv 1

b

i Yes ii No iii Yes iv No

15

a $-2i$ b $-v$ c -4 d 10

e $-27i$ f 16 g $-13 - 7i$ h $-7 - 2i$

Complex conjugates

16 Complex conjugates

17 A real number

18

a 20 b 39 c 26 d 25

e 17 f 3 g 21 h 5

19

a $2 - 3i$ b $-5i + 2$ c $\frac{5i}{6}$ d $2 + 5i$

e $5 + i$ f $2 - i$ g $3 + 2i$ h $-\frac{5}{13} - \frac{12i}{13}$

i $-\frac{2}{17} - \frac{9i}{17}$ j $\frac{2}{5} + \frac{6i}{5}$

20

a 10

b 31

21

a $\dfrac{67-47i}{34}$

b $-7-10i$

22 $\dfrac{16}{5}-5i$

Application

23 $33+50i$